

//PROJECT

```
#include <stdio.h>

void oddEven(int num) {
    if (num % 2 == 0){
        printf("%d is Even.\n", num);}
    else{
        printf("%d is Odd.\n", num);}
}

void positiveNegative(int num) {
    if (num > 0)
        printf("%d is Positive.\n", num);
    else if (num < 0)
        printf("%d is Negative.\n", num);
    else
        printf("%d is Zero.\n", num);
}

void factorial(int num) {
    if (num < 0) {
        printf("Factorial is not defined for negative numbers.\n");
        return;
    }
    long long fact = 1;
    int i;
    for (i = 1; i <= num; i++) {
        fact *= i;
    }
    printf("Factorial of %d is %lld.\n", num, fact);
}

int main() {
    int num;
    char choice;

    while (1) {

        printf("\nMenu:\n");
        printf("A: Find Odd or Even\n");
        printf("B: Find Positive or Negative\n");
        printf("C: Find the Factorial value\n");
        printf("D: Exit\n");
        printf("Enter your choice: ");
        scanf(" %c", &choice);

        if (choice == 'D' || choice == 'd') {
            printf("Exiting the program. Goodbye!\n");
            break;
        }
    }
}
```

```
printf("Enter an integer: ");
scanf("%d", &num);

switch (choice) {
    case 'A':
    case 'a':
        oddEven(num);
        break;
    case 'B':
    case 'b':
        positiveNegative(num);
        break;
    case 'C':
    case 'c':
        factorial(num);
        break;
    default:
        printf("Invalid choice! Please try again.\n");
}

}

return 0;
}
```

OUTPUT:

```
E:\C Programming\project.exe X + v

Menu:
A: Find Odd or Even
B: Find Positive or Negative
C: Find the Factorial value
D: Exit
Enter your choice: a
Enter an integer: 1
1 is Odd.

Menu:
A: Find Odd or Even
B: Find Positive or Negative
C: Find the Factorial value
D: Exit
Enter your choice: B
Enter an integer: 12
12 is Positive.

Menu:
A: Find Odd or Even
B: Find Positive or Negative
C: Find the Factorial value
D: Exit
Enter your choice: c
Enter an integer: 5
Factorial of 5 is 120.

Menu:
A: Find Odd or Even
B: Find Positive or Negative
C: Find the Factorial value
D: Exit
Enter your choice: D
Exiting the program. Goodbye!

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Process exited after 131.8 seconds with return value 0
Press any key to continue . . . |
```